

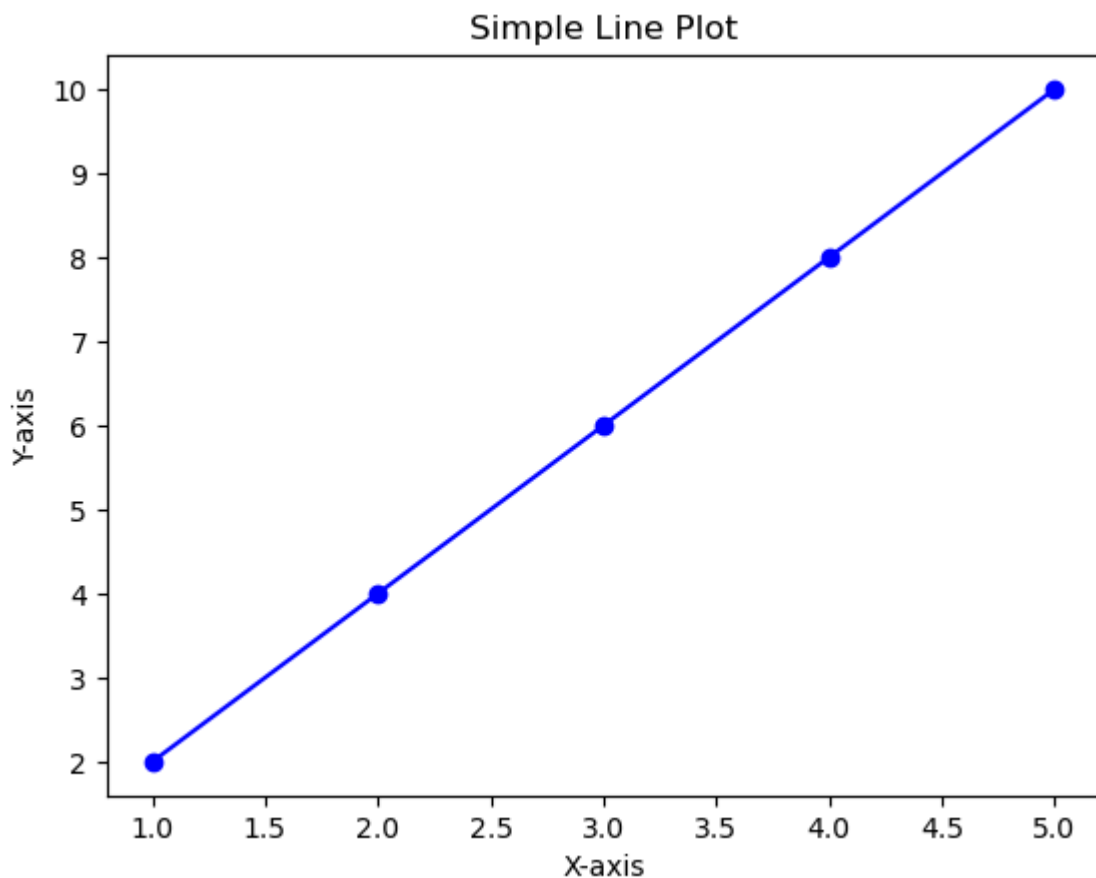
```
In [1]: #Pandas
import matplotlib.pyplot as plt

# Data
x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

# Create plot
plt.plot(x, y, marker='o', linestyle='-', color='blue')

# Add Labels and title
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Simple Line Plot")

# Show plot
plt.show()
```



```
In [4]: ***31-03-2026 working on Pandas wrking on graphs**
```

```
In [8]: import pandas as pd
```

```
In [9]: pd.__version__
```

```
Out[9]: '2.1.4'
```

```
In [12]: df = pd.read_excel(r"C:\Users\Jagatjyoti\Jagat_Code\31-03-2026\data.xlsx")
```

```
In [13]: #df = pd.read_excel("C:/Users/Jagatjyoti/Jagat_Code/31-03-2026/data.xlsx")
```

```
In [14]: print(df.head())
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	

	IncomeGroup
0	High income
1	Low income
2	Upper middle income
3	Upper middle income
4	High income

```
In [15]: len(df)
```

```
Out[15]: 195
```

```
In [16]: print(df.shape)
```

```
(195, 5)
```

```
In [17]: df.columns
```

```
Out[17]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
               'IncomeGroup'],  
              dtype='object')
```

```
In [19]: print(len(df.columns))
```

```
5
```

```
In [20]: df
```

Out[20]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [21]: `df.isnull()`

Out[21]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [22]: `df.isna()`

Out[22]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [23]: `df.isnull().sum()`

```
Out[23]: CountryName      0
CountryCode      0
BirthRate        0
InternetUsers     0
IncomeGroup      0
dtype: int64
```

In [24]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

In [25]: `id(df)`

Out[25]: 2142217550160

In [26]: `df.head()`

Out[26]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [27]: `df.tail()`

Out[27]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

Indexing and slicing in pandas

In [31]: `df[5]`

```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3791, in Index.get_loc(self, key)
    3790 try:
-> 3791     return self._engine.get_loc(casted_key)
    3792 except KeyError as err:

File index.pyx:152, in pandas._libs.index.IndexEngine.get_loc()

File index.pyx:181, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\hashtable_class_helper.pxi:7080, in pandas._libs.hashtable.PyObjectHashTable.get_item()

File pandas\_libs\hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 5

The above exception was the direct cause of the following exception:

KeyError                                Traceback (most recent call last)
Cell In[31], line 1
----> 1 df[5]

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:3893, in DataFrame.__getitem__(self, key)
    3891 if self.columns.nlevels > 1:
    3892     return self._getitem_multilevel(key)
-> 3893 indexer = self.columns.get_loc(key)
    3894 if is_integer(indexer):
    3895     indexer = [indexer]

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3798, in Index.get_loc(self, key)
    3793     if isinstance(casted_key, slice) or (
    3794         isinstance(casted_key, abc.Iterable)
    3795         and any(isinstance(x, slice) for x in casted_key)
    3796     ):
    3797         raise InvalidIndexError(key)
-> 3798     raise KeyError(key) from err
    3799 except TypeError:
    3800     # If we have a listlike key, _check_indexing_error will raise
    3801     # InvalidIndexError. Otherwise we fall through and re-raise
    3802     # the TypeError.
    3803     self._check_indexing_error(key)

KeyError: 5

```

```
In [32]: df[:5]
```

Out[32]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [33]:

```
df[5,10]
```

```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3791, in Index.get_loc(self, key)
    3790 try:
-> 3791     return self._engine.get_loc(casted_key)
    3792 except KeyError as err:

File index.pyx:152, in pandas._libs.index.IndexEngine.get_loc()

File index.pyx:181, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\hashtable_class_helper.pxi:7080, in pandas._libs.hashtable.PyObjectHashTable.get_item()

File pandas\_libs\hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: (5, 10)

The above exception was the direct cause of the following exception:

KeyError                                Traceback (most recent call last)
Cell In[33], line 1
----> 1 df[5,10]

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:3893, in DataFrame.__getitem__(self, key)
    3891 if self.columns.nlevels > 1:
    3892     return self._getitem_multilevel(key)
-> 3893 indexer = self.columns.get_loc(key)
    3894 if is_integer(indexer):
    3895     indexer = [indexer]

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3798, in Index.get_loc(self, key)
    3793     if isinstance(casted_key, slice) or (
    3794         isinstance(casted_key, abc.Iterable)
    3795         and any(isinstance(x, slice) for x in casted_key)
    3796     ):
    3797         raise InvalidIndexError(key)
-> 3798     raise KeyError(key) from err
    3799 except TypeError:
    3800     # If we have a listlike key, _check_indexing_error will raise
    3801     # InvalidIndexError. Otherwise we fall through and re-raise
    3802     # the TypeError.
    3803     self._check_indexing_error(key)

KeyError: (5, 10)

```

```
In [35]: df[5:10]
```


Out[35]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income

In [36]: df[3:]

Out[36]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
6	Armenia	ARM	13.308	41.9	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

192 rows × 5 columns

In [37]: df[:34]

Out[37]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90000	High income
1	Afghanistan	AFG	35.253	5.90000	Low income
2	Angola	AGO	45.985	19.10000	Upper middle income
3	Albania	ALB	12.877	57.20000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.00000	High income
5	Argentina	ARG	17.716	59.90000	High income
6	Armenia	ARM	13.308	41.90000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.40000	High income
8	Australia	AUS	13.200	83.00000	High income
9	Austria	AUT	9.400	80.61880	High income
10	Azerbaijan	AZE	18.300	58.70000	Upper middle income
11	Burundi	BDI	44.151	1.30000	Low income
12	Belgium	BEL	11.200	82.17020	High income
13	Benin	BEN	36.440	4.90000	Low income
14	Burkina Faso	BFA	40.551	9.10000	Low income
15	Bangladesh	BGD	20.142	6.63000	Lower middle income
16	Bulgaria	BGR	9.200	53.06150	Upper middle income
17	Bahrain	BHR	15.040	90.00004	High income
18	Bahamas, The	BHS	15.339	72.00000	High income
19	Bosnia and Herzegovina	BIH	9.062	57.79000	Upper middle income
20	Belarus	BLR	12.500	54.17000	Upper middle income
21	Belize	BLZ	23.092	33.60000	Upper middle income
22	Bermuda	BMU	10.400	95.30000	High income
23	Bolivia	BOL	24.236	36.94000	Lower middle income
24	Brazil	BRA	14.931	51.04000	Upper middle income
25	Barbados	BRB	12.188	73.00000	High income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
26	Brunei Darussalam	BRN	16.405	64.50000	High income
27	Bhutan	BTN	18.134	29.90000	Lower middle income
28	Botswana	BWA	25.267	15.00000	Upper middle income
29	Central African Republic	CAF	34.076	3.50000	Low income
30	Canada	CAN	10.900	85.80000	High income
31	Switzerland	CHE	10.200	86.34000	High income
32	Chile	CHL	13.385	66.50000	High income
33	China	CHN	12.100	45.80000	Upper middle income

In [38]: `df[0:200:25]`

Out[38]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
25	Barbados	BRB	12.188	73.000000	High income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
75	Honduras	HND	21.593	17.800000	Lower middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
125	Namibia	NAM	29.937	13.900000	Upper middle income
150	Sudan	SDN	33.477	22.700000	Lower middle income
175	Tunisia	TUN	19.800	43.800000	Upper middle income

In [39]: `df[:, :-1]`

Out[39]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

In [40]: `df[:, -10]`

Out[40]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
184	Venezuela, RB	VEN	19.842	54.9000	High income
174	Trinidad and Tobago	TTO	14.590	63.8000	High income
164	Swaziland	SWZ	30.093	24.7000	Lower middle income
154	Sierra Leone	SLE	36.729	1.7000	Low income
144	French Polynesia	PYF	16.393	56.8000	High income
134	Oman	OMN	20.419	66.4500	High income
124	Malaysia	MYS	16.805	66.9700	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.2400	Upper middle income
104	Lesotho	LSO	28.738	5.0000	Lower middle income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
84	Iceland	ISL	13.400	96.5468	High income
74	Hong Kong SAR, China	HKG	7.900	74.2000	High income
64	Guinea	GIN	37.337	1.6000	Low income
54	Estonia	EST	10.300	79.4000	High income
44	Czech Republic	CZE	10.200	74.1104	High income
34	Cote d'Ivoire	CIV	37.320	8.4000	Lower middle income
24	Brazil	BRA	14.931	51.0400	Upper middle income
14	Burkina Faso	BFA	40.551	9.1000	Low income
4	United Arab Emirates	ARE	11.044	88.0000	High income

In [41]: df[:, :-50]

Out[41]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
144	French Polynesia	PYF	16.393	56.8000	High income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
44	Czech Republic	CZE	10.200	74.1104	High income

```
In [42]: df[::50]
```

Out[42]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
150	Sudan	SDN	33.477	22.700000	Lower middle income

```
In [ ]:
```